



MZUZU UNIVERSITY

FACULTY OF ENVIRONMENTAL SCIENCES

DEPARTMENT OF AGRI-SCIENCES

INTRODUCTION TO STATISTICAL ANALYSIS (STAT 2301)

END OF SEMESTER EXAMINATION

Date: 30th August, 2021

Time: 14:00-17:00 (3 hours)

Answer all questions

Question 1

What are basic principles of experimental designs?

25 marks

Question 2

An experiment to determine the effect of 5 treatments (A, B, C, D, E) was carried out using a Complete Randomized Design and the results are given in the table below. Conduct an Analysis of Variance.

25 marks

		Treatment				
		A	B	C	D	E
Replicate	1	42.2	28.4	18.8	41.5	33.0
	2	34.9	28.0	19.5	36.3	26
	3	29.7	22.8	13.1	31.7	30.6
	4	30	18.5	10.1	31	25
	5	41	19.4	14	28.2	22

Question 3

Descriptive statistics are used for organizing and summarizing data. Discuss.

25marks

Question 4

Calculate the following statistics for **each class** given below:

- | | |
|-------------------------|---------|
| i. Mean | 3 marks |
| ii. Median | 3 marks |
| iii. Mode | 3 marks |
| iv. Range | 3 marks |
| v. Interquartile range | 4 marks |
| vi. Variance | 6 marks |
| vii. Standard Deviation | 3 marks |

Class A	Class B
102	87
128	109
98	111
140	103
98	162
110	109
115	120
109	93
106	80
89	96
119	131
98	127
56	105